

# Dinmukhamet Murat

Robotics Software Engineer | 3D Perception | Real-time Control | ROS 2

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## Summary

Robotics software engineer in industrial **3D perception** and **real-time control**. Owns the production scan-to-CAD, inspection, and assembly-planning stack for ABB robotic welding cells at Quant Robotics, cutting robot placement error from 3 mm to under 1 mm. Deep point-cloud work in Open3D/ICP, plus NMPC motion control across ROS 2, Nav2, and MoveIt2.

## Technical Skills

**3D Vision & Point Clouds:** Open3D, ICP, RANSAC, FPFH, DBSCAN, raycasting, scan-to-CAD matching, PyVista, Trimesh  
**Robotics & ROS:** ROS 2 (Humble / Jazzy), MoveIt2, ABB RAPID, Nav2 (SLAM), Gazebo, RViz, URDF / SRDF / xacro  
**Control & Calibration:** MPC / NMPC, ACADOS, trajectory optimization, SSM / pHRI safety, TCP & scanner calibration  
**ML & Edge CV:** YOLOv8 / YOLOv11 (detection, 3D bboxes), OpenVINO, ONNX, OpenCV, NumPy, SciPy  
**Programming & Infrastructure:** Python (3.8-3.12), Bash, CMake, Docker, Git / GitLab, MongoDB, Linux, AWS

## Experience

### Production 3D Perception & Scan-to-CAD for Robotic Welding Cells

June 2025 - Present

*Robotics Software Engineer, Quant Robotics*

- Rebuilt the Open3D scan-to-CAD registration pipeline, cutting ABB placement error from 3 mm to **under 1 mm** - meeting the required 1 mm assembly tolerance that manual welding (error up to 3 mm) cannot reliably hold.
- Own the production 3D scanning, registration, and inspection stack for the EVA robotic welding cells: modular Python behind one Docker entrypoint, orchestrating an ABB IRB6700 and Gocator laser scanner via MongoDB.
- Introduced raycasting-based beam registration (new to the codebase), simulating scanner-visible CAD point clouds and aligning scanned beams to CAD through staged ICP for downstream assembly planning.
- Eliminated mis-identification of look-alike parts (same shape, different thickness) via thickness grouping and edge-count pre-filtering, preventing wrong parts from being welded into an assembly.
- Parallelized independent beam end-face and on-beam part registration, cutting beam matching from **80 s to 15 s (5x)** and table matching from 60-75 s to 15-20 s.
- Cut install-and-weld planning from **160 s to 30-55 s** in production, so it finishes within the robot's scanner-to-magnet tool change and removes the multi-minute idle wait between scan and assembly.
- Added optimal-trajectory selection for the welder's retreat moves: the planner builds several candidate joint-space plans and runs the lowest-motion one, so the welder stays near the part-placement zone.
- Developed robot-to-robot part-position handoff in ABB RAPID, validated on the production cell: the IRB6700 serves the measured part correction over a socket and the IRB2600 applies it to its weld targets.
- Automated TCP calibration for magnets and the table, replacing a 10-minute operator-dependent routine with a hands-off step that completes in **under 5 s** and holds sub-millimetre scanner alignment.

### Real-time NMPC Motion Control for Mobile Robots

May 2024 - June 2025

*Research Assistant, Nazarbayev University*

- Developed a custom NMPC local controller integrated into Nav2 for a Clearpath Jackal UGV (ROS 2 Humble / Gazebo), solving the optimization in **50-70 ms** per cycle with the ACADOS solver on a skid-steer model.
- Implemented human-aware velocity adaptation under Speed-and-Separation Monitoring constraints, integrating YOLOv8-nano 3D person detection (yolo\_ros) into the control loop at 10 Hz with under 3% CPU overhead.

### Edge Computer-Vision Performance Optimization

Feb 2025 - Apr 2025

*Edge Video Systems Engineer, FOQUS (Remote)*

- Improved YOLO-nano inference throughput from 20 FPS (OpenVINO IR baseline) to **33-37 FPS** (ONNX and OpenVINO) by benchmarking YOLOv5/v8/v11 and tuning export, device fallback, and frame-level logging.

## Projects

- virtual-pointcloud-scanner** - simulates industrial laser-scanner capture from 36 viewpoints via Open3D raycasting (tensor API) with a pinhole model, handling occlusion that naive mesh sampling misses.
- ros\_moveit\_pnp** (ROS 2 Jazzy, MoveIt2, UR-5; in development) - a UR-5 picks boxes from a conveyor and arranges them into target shapes: cube, tower, or pyramid.

## Education

B.S., Robotics Engineering

June 2025

Nazarbayev University, Astana, Kazakhstan